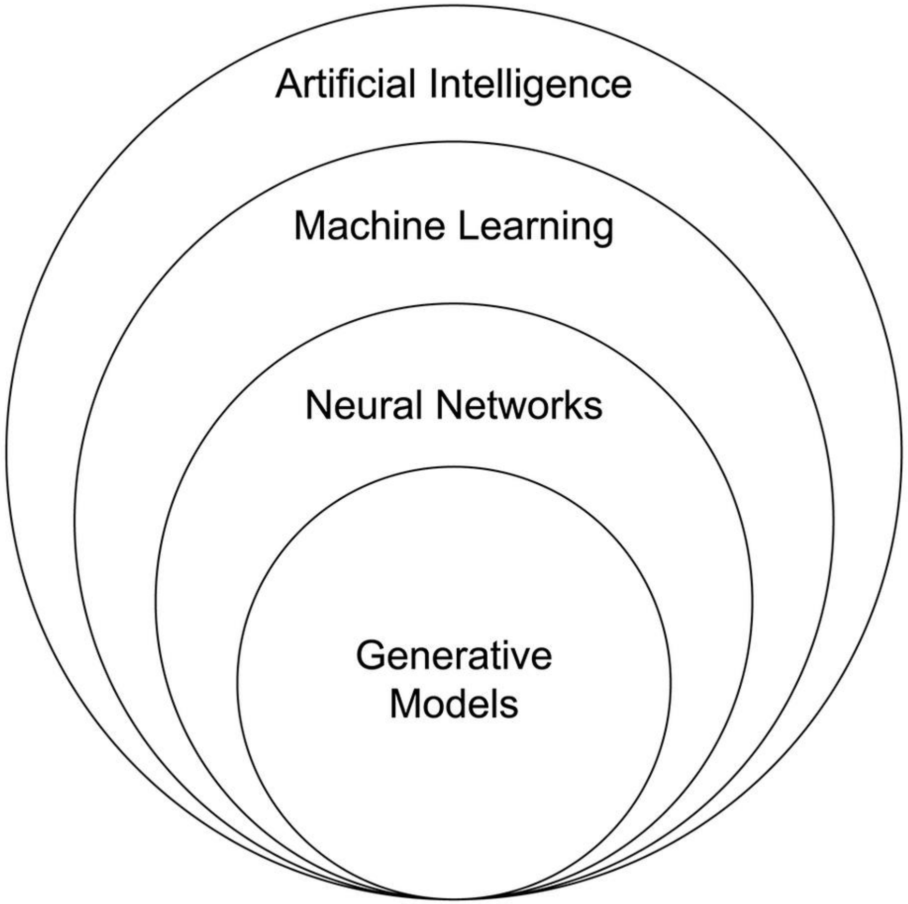
Machine Learning 1

Terminology

- AI
- ML
- Deep learning
- Statistical learning



https://commons.wikimedia.org/wiki/File:AI_relation_to_Generative_Models_subset,_venn_diagram.png

High-dimension datasets (n << p)

- SNP arrays
- RNA-seq
- Single-cell RNA-seq
- Methylation/epigenetics
- Proteomics
- Metabolomics
- Images
- Text/language
- …

2D plots are not enough

- Try exercises 1 and 2

The curse of dimensionality

Not just "there are a lot of variables, and we don't know which are important"

- Distance concentration
  - Points become ~equidistant
  - → "Nearest neighbor" means less
- Volume
  - Grows exponentially with p
  - → Data extremely sparse
  - → Huge sample size needed to cover the space
- Most of the volume is near the boundary
  - Very little interior
  - "Distance from center" means less
- Projections are deceptive
  - Lower-dimensional subsets will *appear* to have structure even if they're pure noise
- Statistical problems (more variables than datapoints)
  - → need for feature selection/extraction
  - → easy to find "perfect fit" based on noise alone (overfitting)

See Example 3

What kind of structure?

ML methods assume structure.

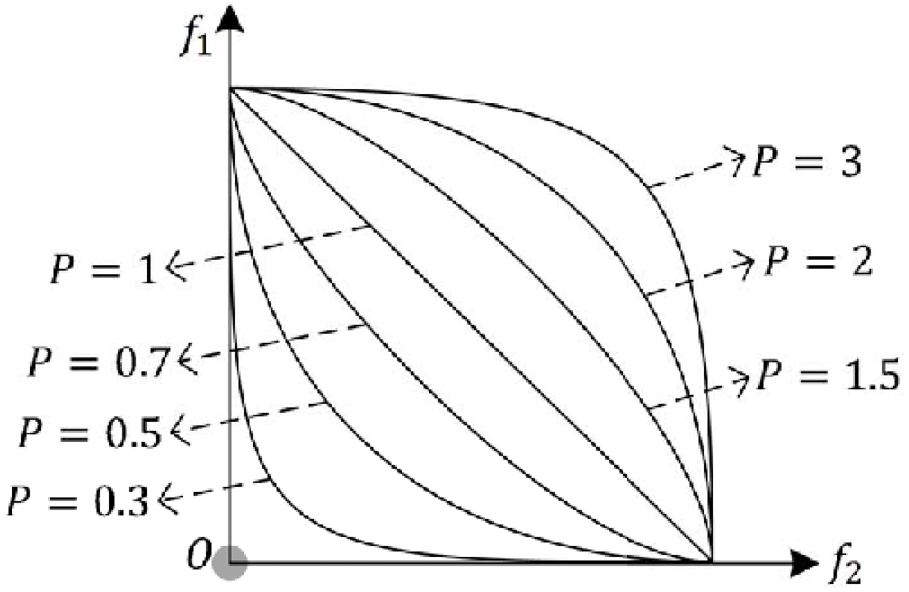
It’s up to you to ask the right question and choose the right tool.

- Are there clusters/groups?
- Does variation exist globally along smooth gradients?
- Is there predictive structure (i.e., can some variables predict others?)
- Are there local structures?
- Is there hierarchical structure?
- Are there networks/relational structures?
- How complex is the structure? What is the intrinsic dimensionality?

Supervised vs. unsupervised ML answer different questions

Type of ML	Kind of question	You are given	Goal	Assumptions
Supervised	Is there a mapping from X→Y?	• Input variables (X) • Target/outcome/labels (Y)	• Predict Y • Estimate relationships between X and Y	• The predictive structure (model type) • Signal in X could explain Y
Unsupervised	What structure exists in X?	• Only the data (X) • No outcome/labels	• Describe variation • Find groups, dimensions, gradients, etc.	• Continuous or discrete structure • Local or relational structure

How to measure distance



Xu et al. 2018 IEEE Transactions on Cybernetics 49(11)

- Minkowski distance = $d_p(x, y) = \sum_{i=1}^{n} |x_i - y_i|^p$
    - Euclidean distance (p=2)
    - Manhattan Distance (p=1)
    - More generally, p >= 1
- Correlation distance
- Mahalanobis distance (analogous to a higher-dimensional z-score)

$$d_M(\vec{x}, \vec{y}; Q) = \sqrt{(\vec{x} - \vec{y})^\mathsf{T} \mathbf{\Sigma}^{-1} (\vec{x} - \vec{y})}.$$

Euclidean distance example

Euclidean Distance b/w observation A and B in 2-D space (Feature 1 and 2)
= $\sqrt{(60-45)^2+(55-40)^2}$

Observation\features	Feature 1	Feature 2	Feature 3	Feature 4	Feature 5
Observation A	45	40	35	35	56
Observation B	60	55	50	50	71

Euclidean distance example

Euclidean Distance b/w observation A and B in 2-D space (Feature 1 and 2)

= $\sqrt{(60-45)^2+(55-40)^2}$

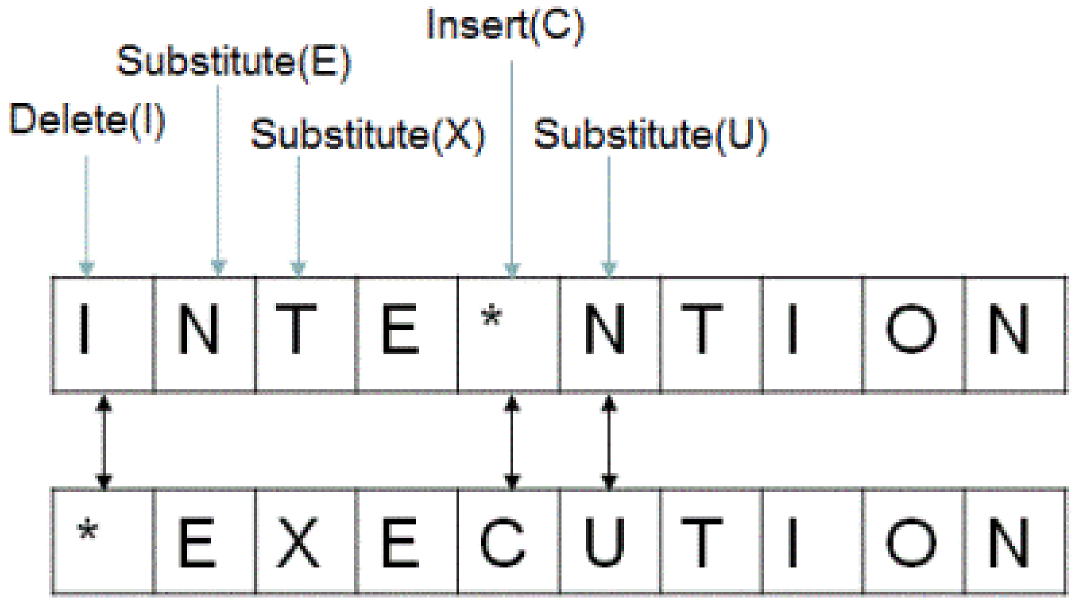
- Euclidean Distance b/w observation A and B in 5-D space (Feature 1~5)

= $\sqrt{(60-45)^2+(55-40)^2+(50-35)^2+(50-35)^2+(71-56)^2}$

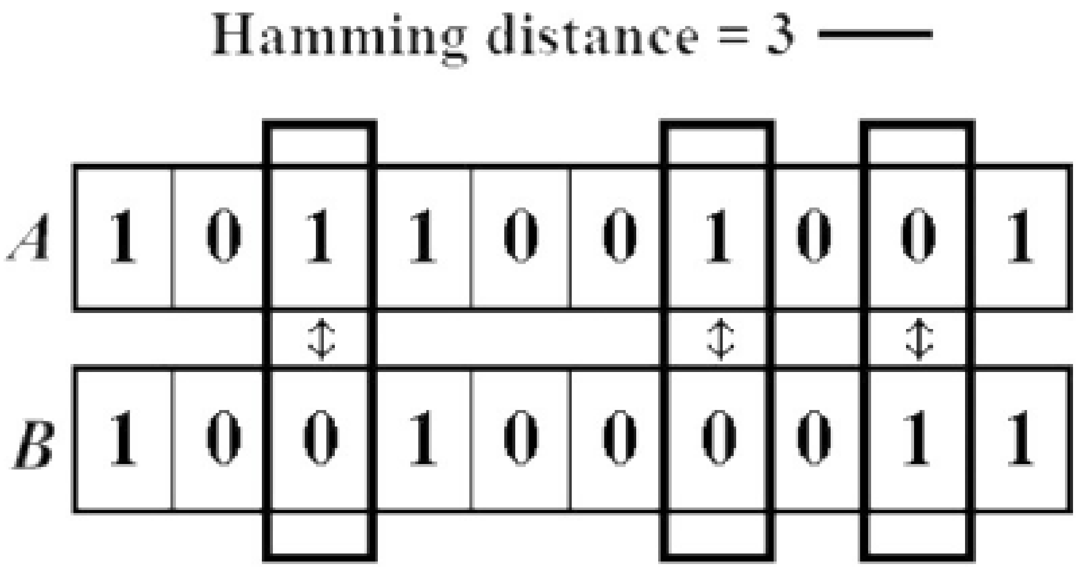
Observation\features	Feature 1	Feature 2	Feature 3	Feature 4	Feature 5
Observation A	45	40	35	35	56
Observation B	60	55	50	50	71

Other metrics

- Hamming distance
- Levenshtein distance



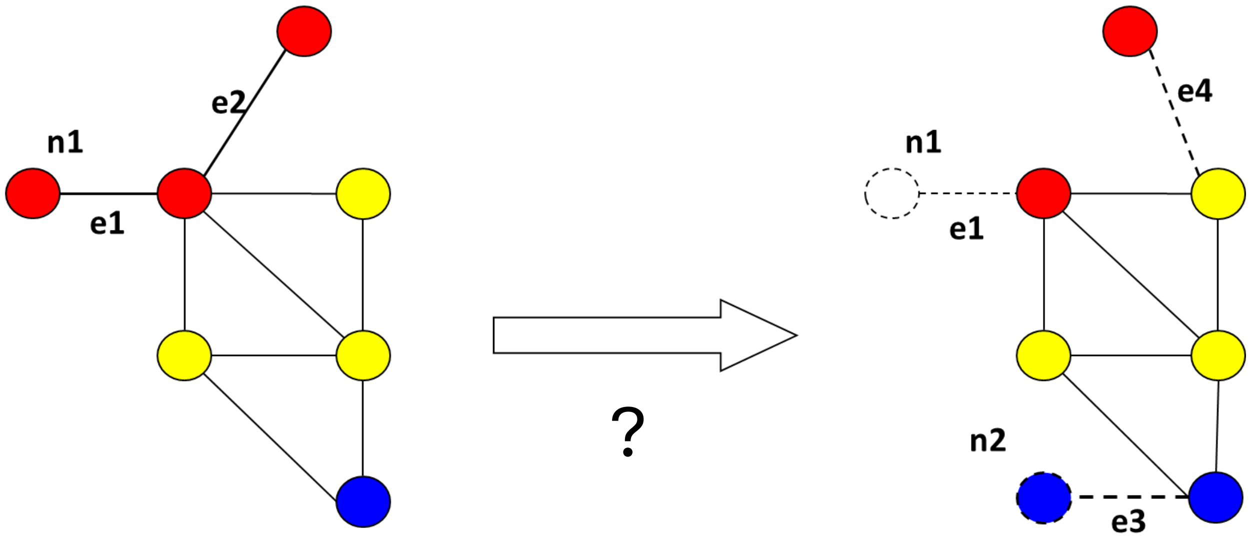
https://viblo.asia/p/edit-distance-L4x5xdQ15BM



Ishengoma, DOI: 10.5815/ijieeb.2014.06.08

Other distances (con't.)

- The Edit Distance (Levenshtein distance) between two strings, str1 and str2 is the number of the number of operations required to transform str2 into str1
- Requires a set of operators
- E. g., graph edit distance with operators for
  - Edge add/delete
  - Node add/delete
- E. g., distance between two genotype sequences of the same length



https://viblo.asia/p/edit-distance-L4x5xdQ15BM

Unsupervised ML

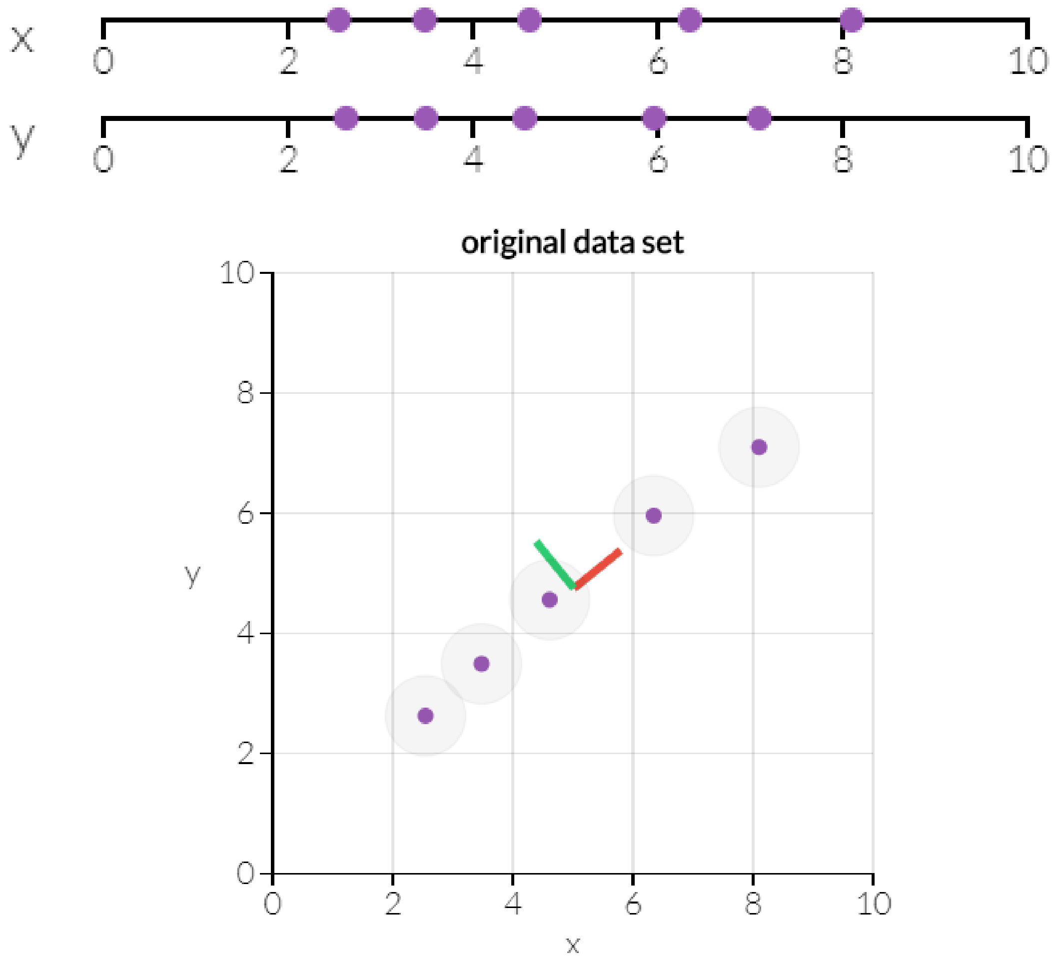
Main kinds of techniques, and some specific algorithms

- Dimension reduction
  - PCA, MDS, etc.
  - UMAP/t-SNE
- Clustering
  - Hierarchical clustering
  - k-means

Principal components analysis for dimension reduction

- How to find 2D (two variables) space in which information among data points are mostly kept?
- With variance in data as information, PCA can find a 1-d line preserving most information.

5 samples on a 2-d space (X and Y)

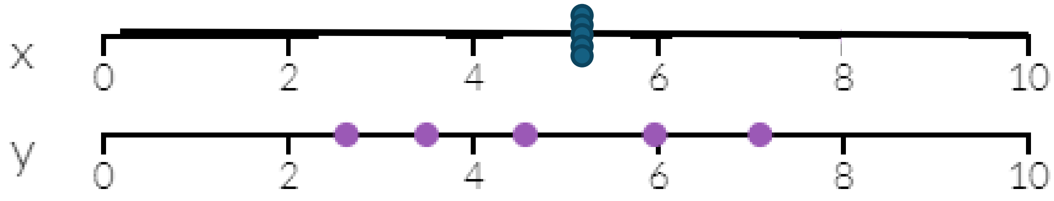


http://setosa.io/ev/principal-component-

Principal components maximize variance

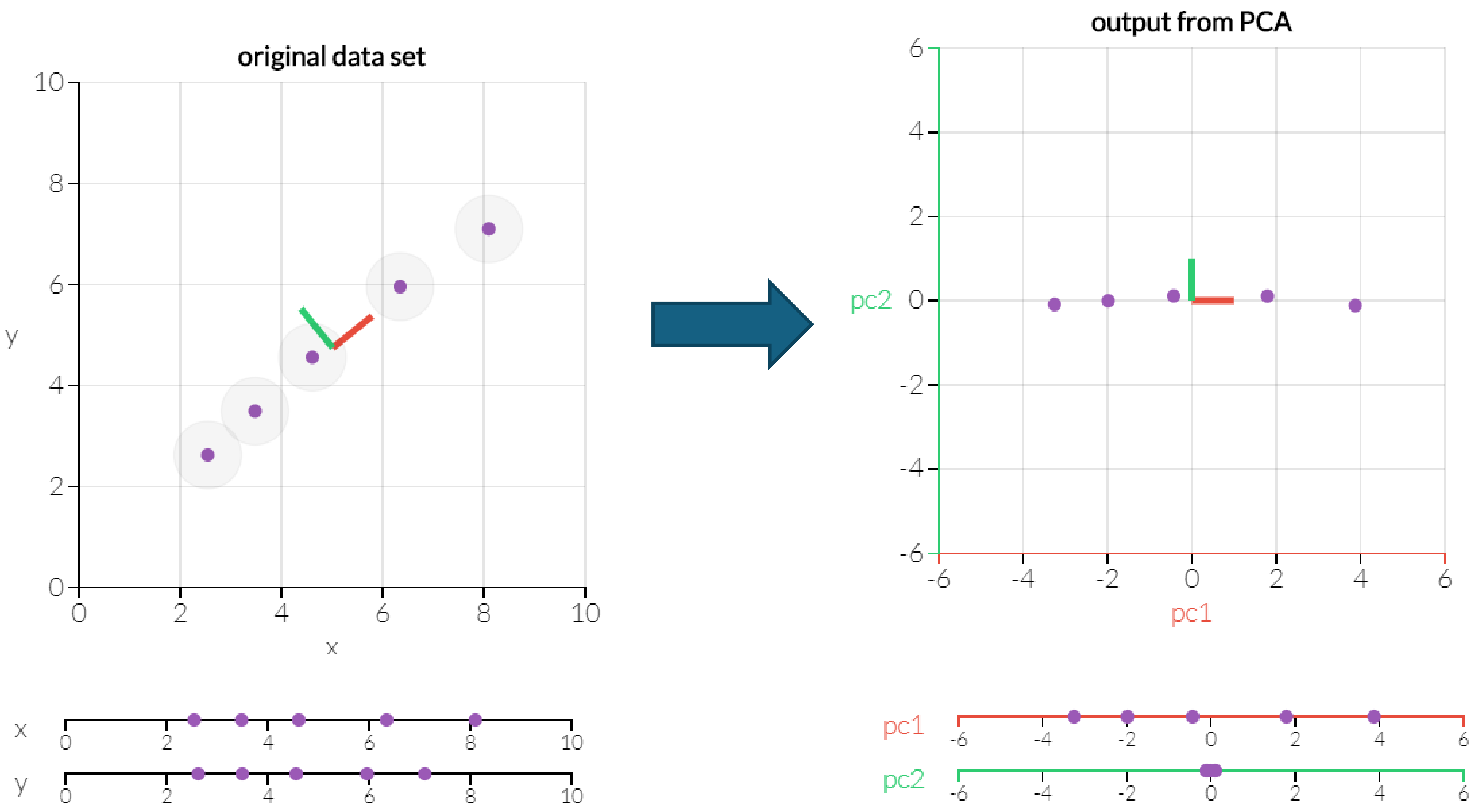
- How to find 2D (two variables) space in which information among data points are mostly kept?
- Variance in data can be taken as information

5 samples on a 2-d space (X and Y)



Adapted from http://setosa.io/ev/principal-component-analysis/

Principal components maximize variance

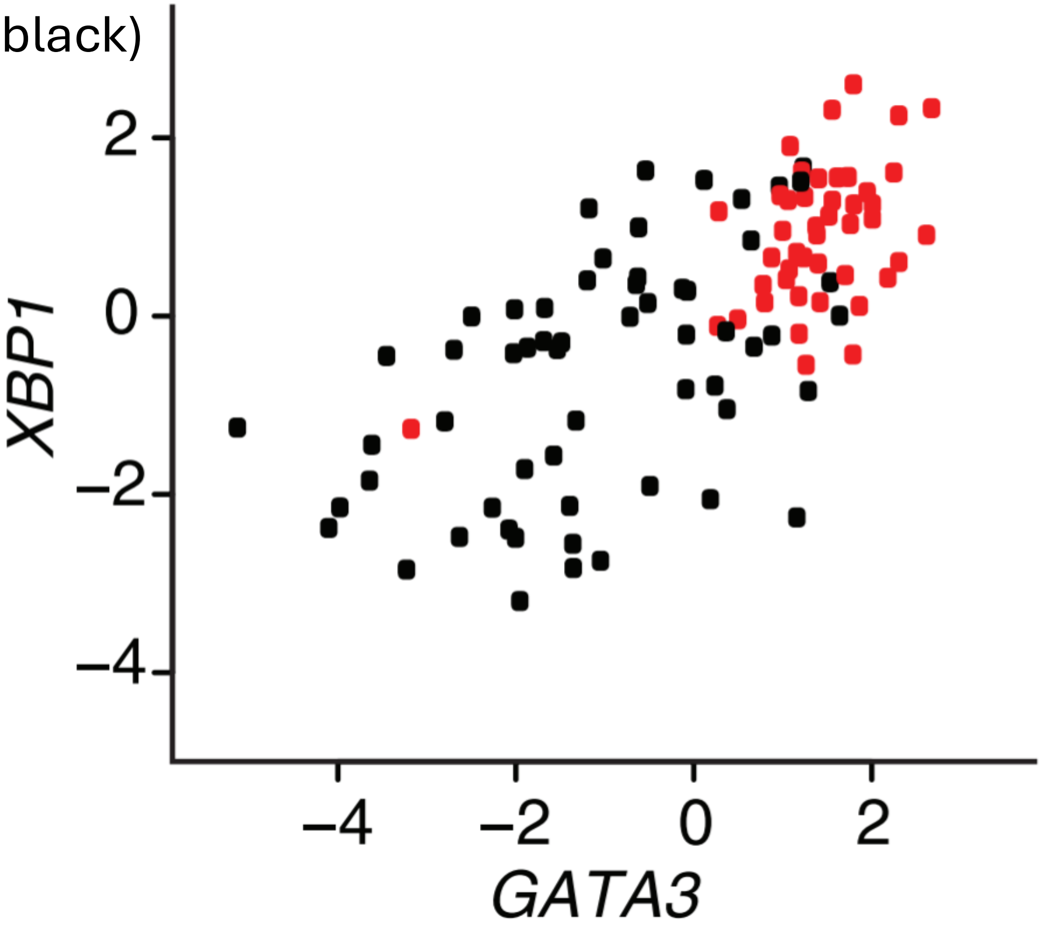


- Can identify two axes that maximize variance in high dimension

http://setosa.io/ev/principal-component-

Principal components in biomedical research

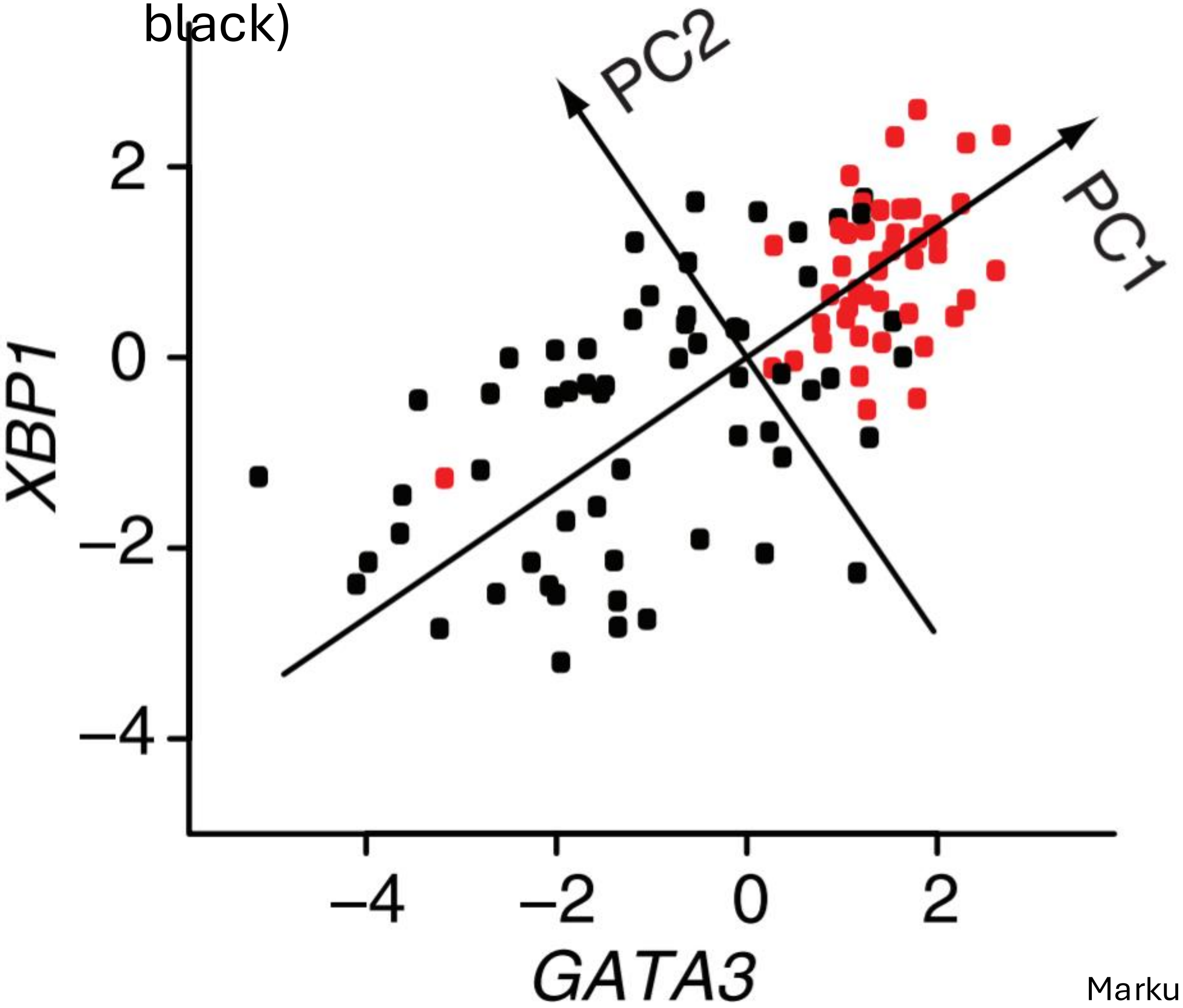
Breast cancer samples (ER+ in red and ER- in black)



Markus, Nat. Biotechnology, Vol. 26 Num. 3,

Principal components in biomedical research

Breast cancer samples (ER+ in red and ER- in black)

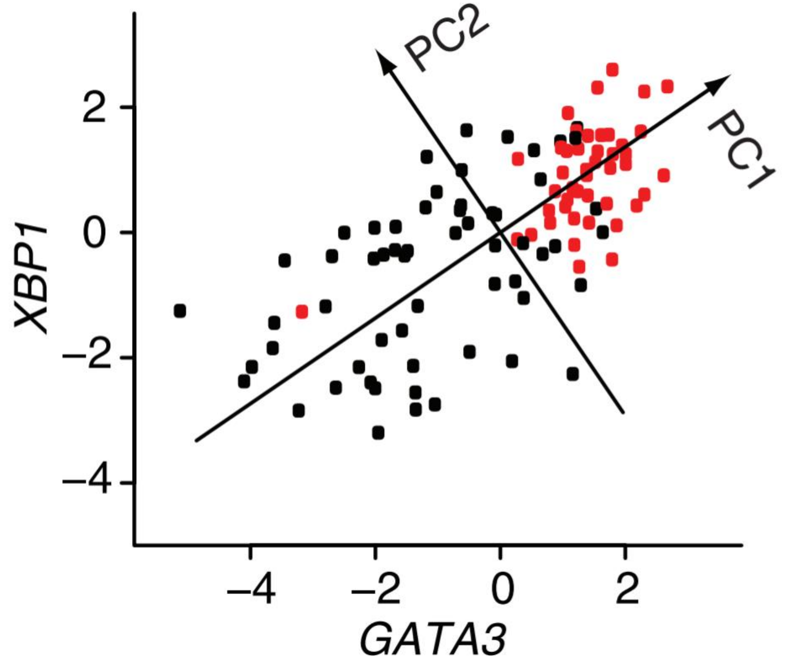


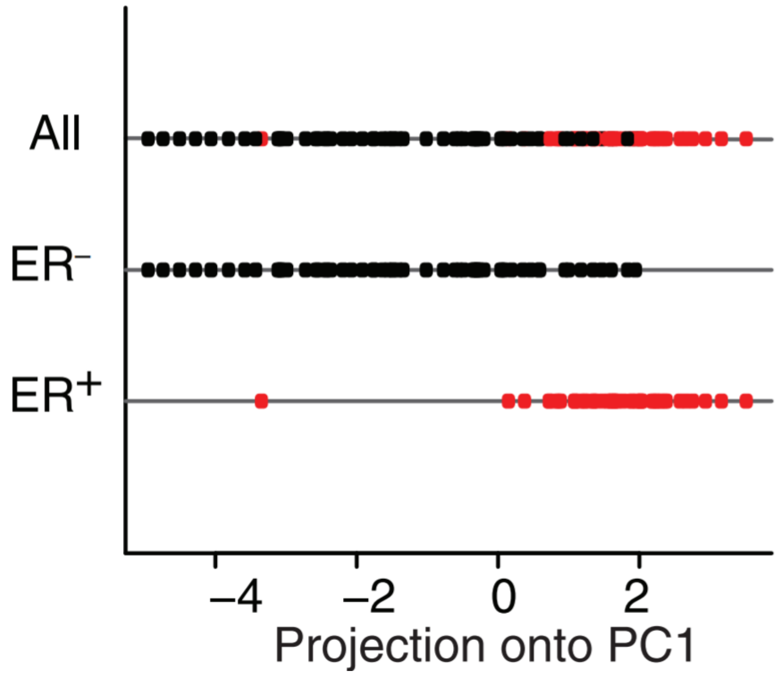
Markus, Nat. Biotechnology, Vol. 26 Num. 3,

Principal components in biomedical research

Things to analyze in PCA

1. Their projection to PCAs: PC1 well separates the grouping of your interest



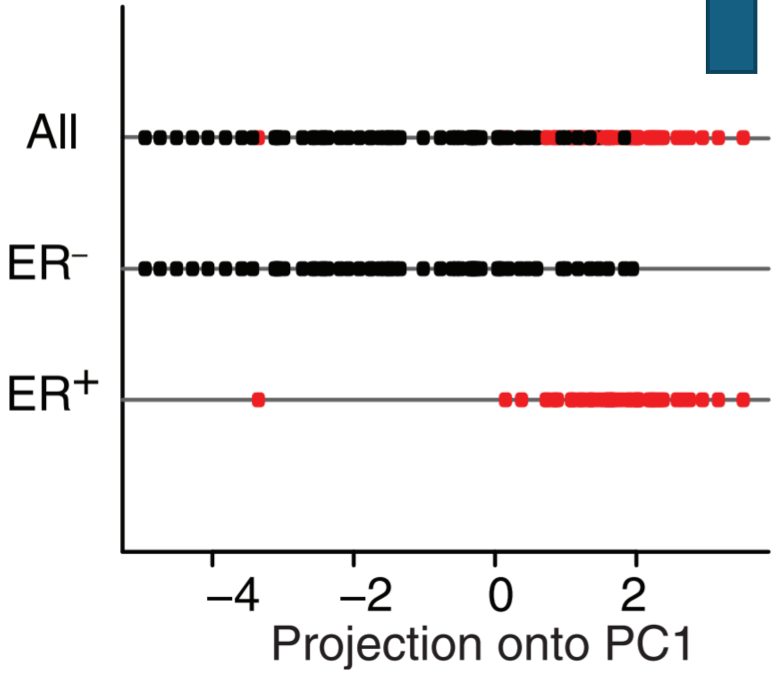


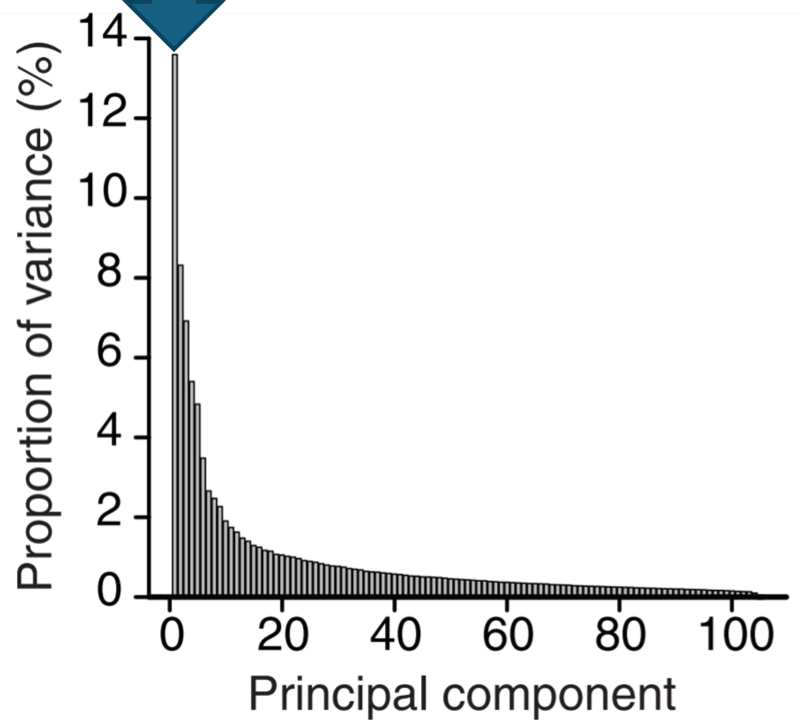
Markus, Nat. Biotechnology, Vol. 26 Num. 3,

Principal components in biomedical research

Some things to check in PCA

1. Projection of data onto PCs
2. Proportion of variance per PC



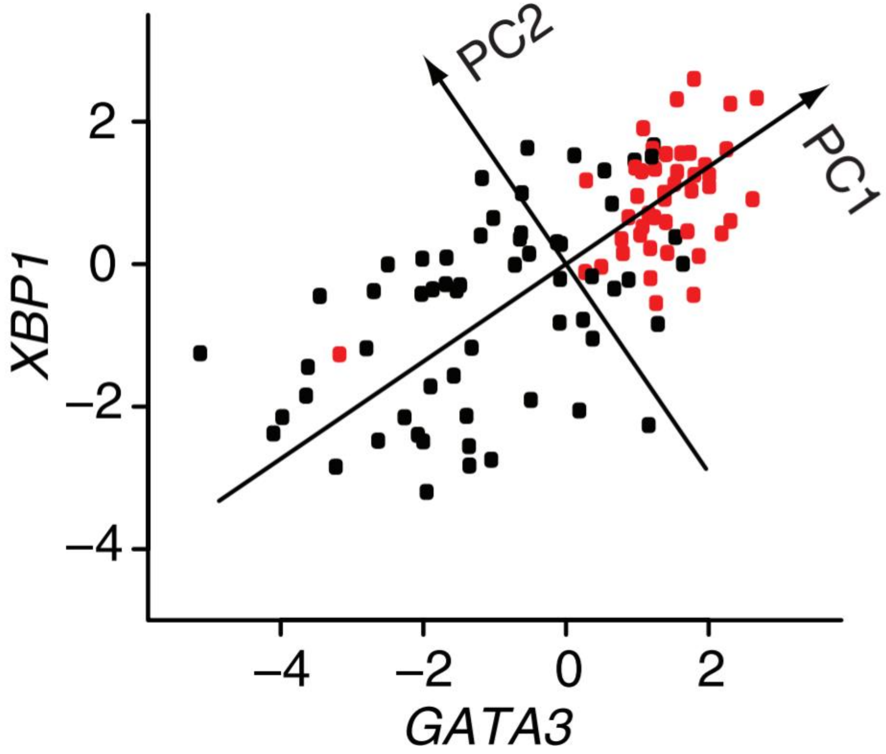


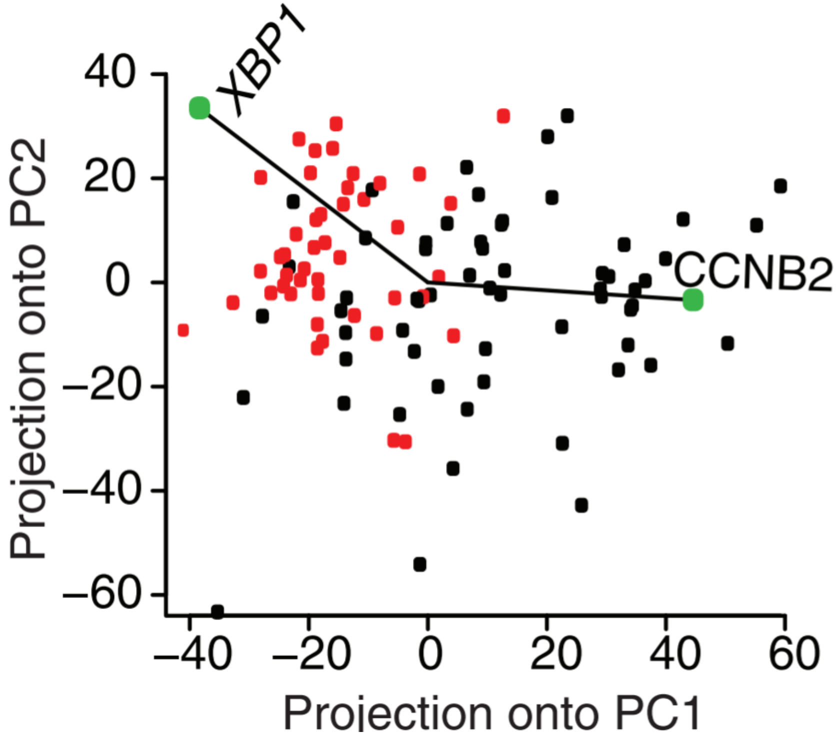
Markus, Nat. Biotechnology, Vol. 26 Num. 3,

Principal components in biomedical research

Things to check in PCA

1. Projection of data onto PCs
2. Proportion of variance per PC
3. Weights of your genes on the PCs



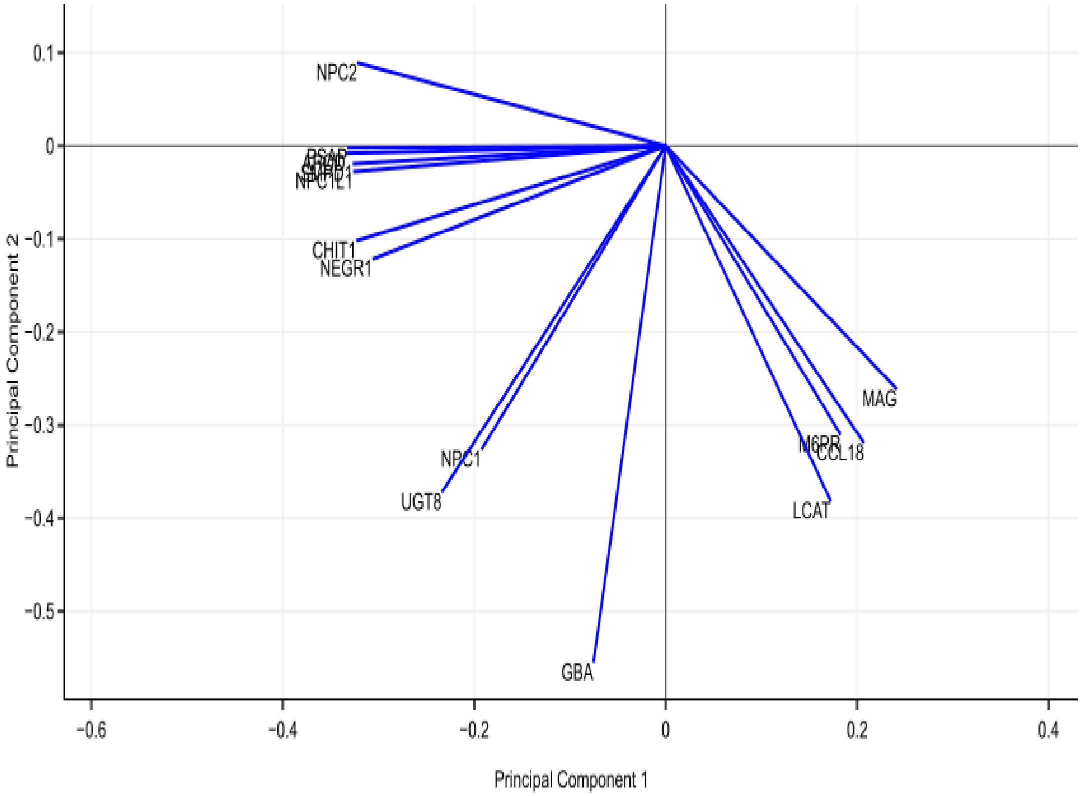


Markus, Nat. Biotechnology, Vol. 26 Num. 3,

A loading plot shows how strongly each dimension influences a principal component.

A loading plot shows

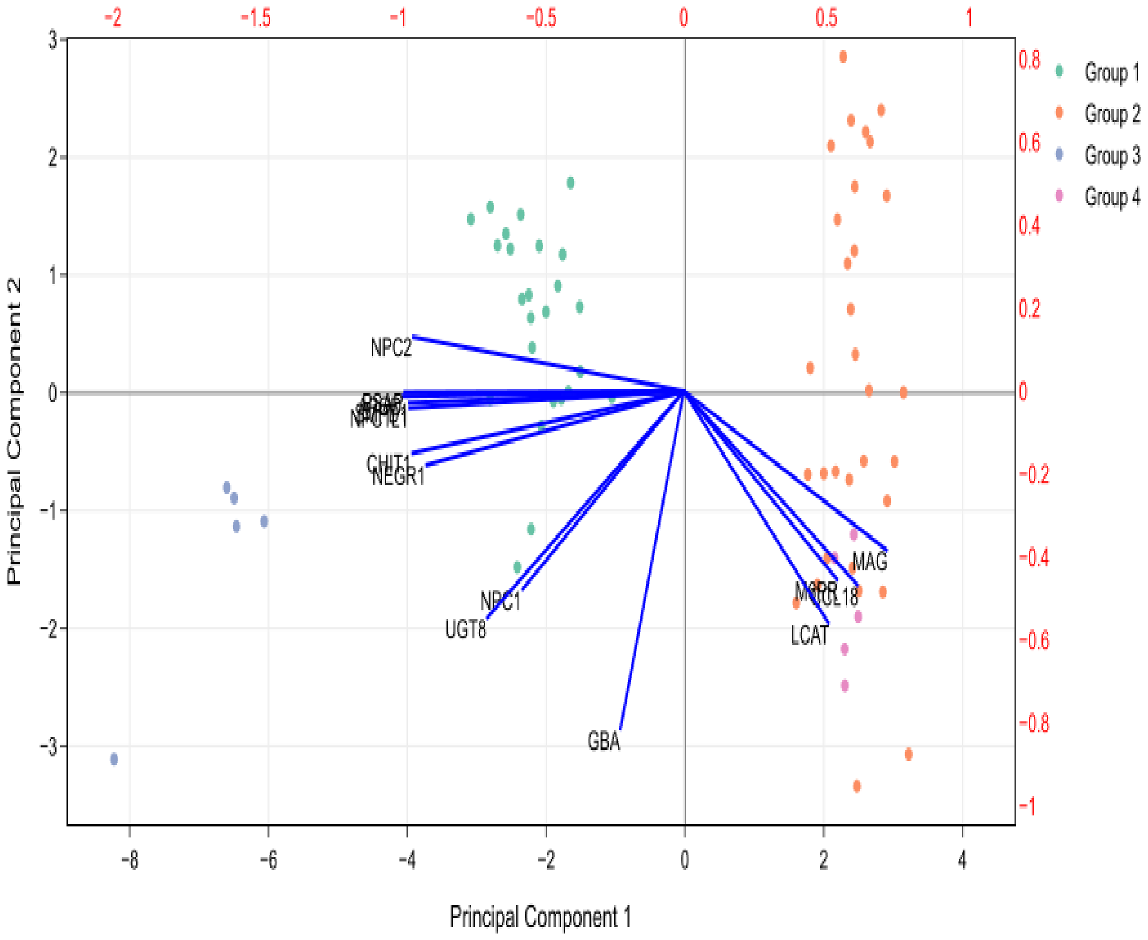
- Their projected values on each PC show how much weight they have on that PC
  - e. g., NPC2 and MAG influence PC1
  - e. g., GBA influences PC2
- The angles b/w variables show how they correlated with each other
  - e. g., APOD and PSAP positively correlated
  - NPC2 and GBA not likely to be correlated
  - NPC2 and MAG negatively correlated



http://setosa.io/ev/principal-component-analysis/

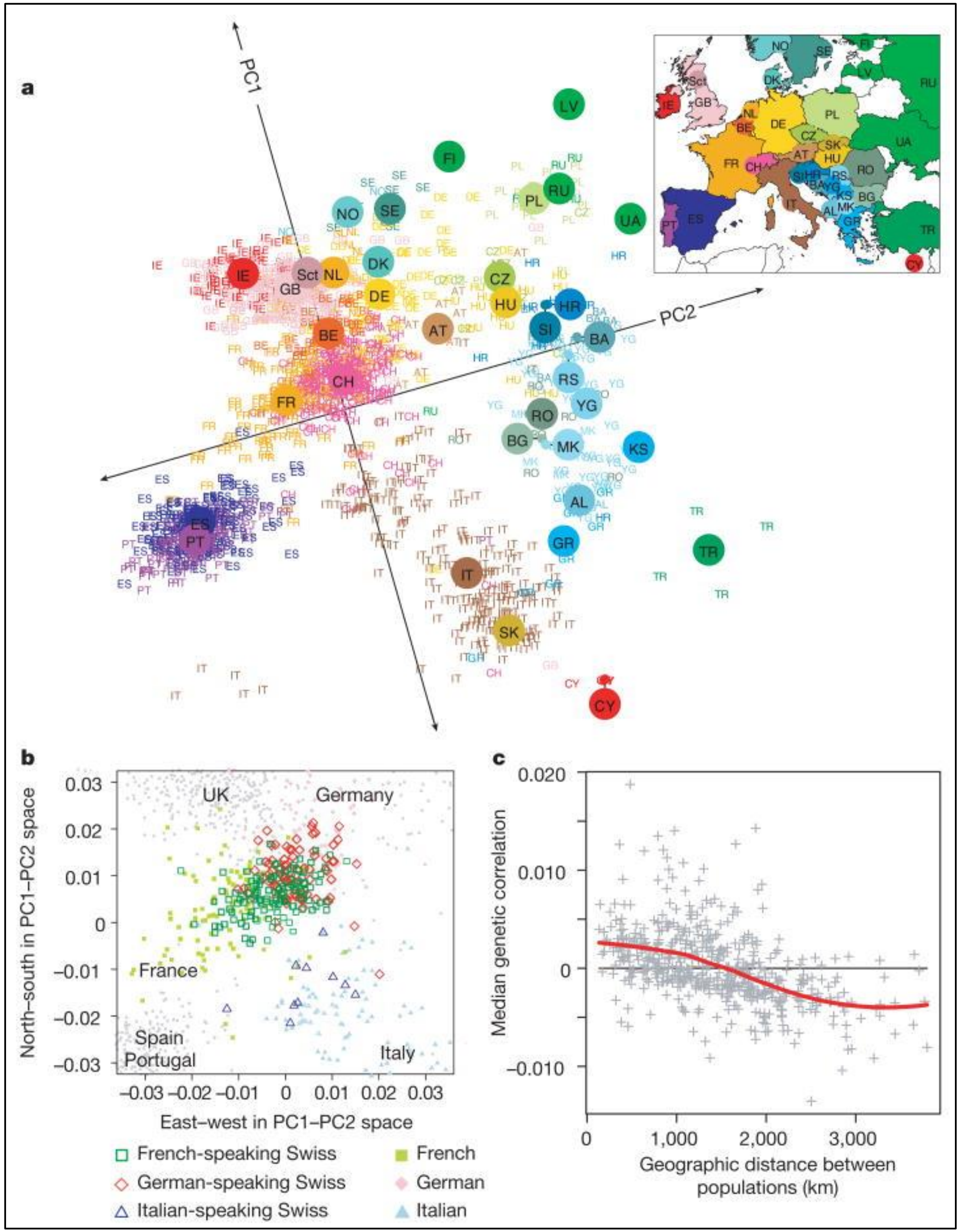
PCA biplot = PCA score plot + loading plot

- Bottom axis: PC1 score
- Left axis: PC2 score
- Top axis: loadings on PC1
- Right axis: loadings on PC2



http://setosa.io/ev/principal-component-analysis/

Example of PCA in genetics: principal components of ancestry



Limitations of PCA

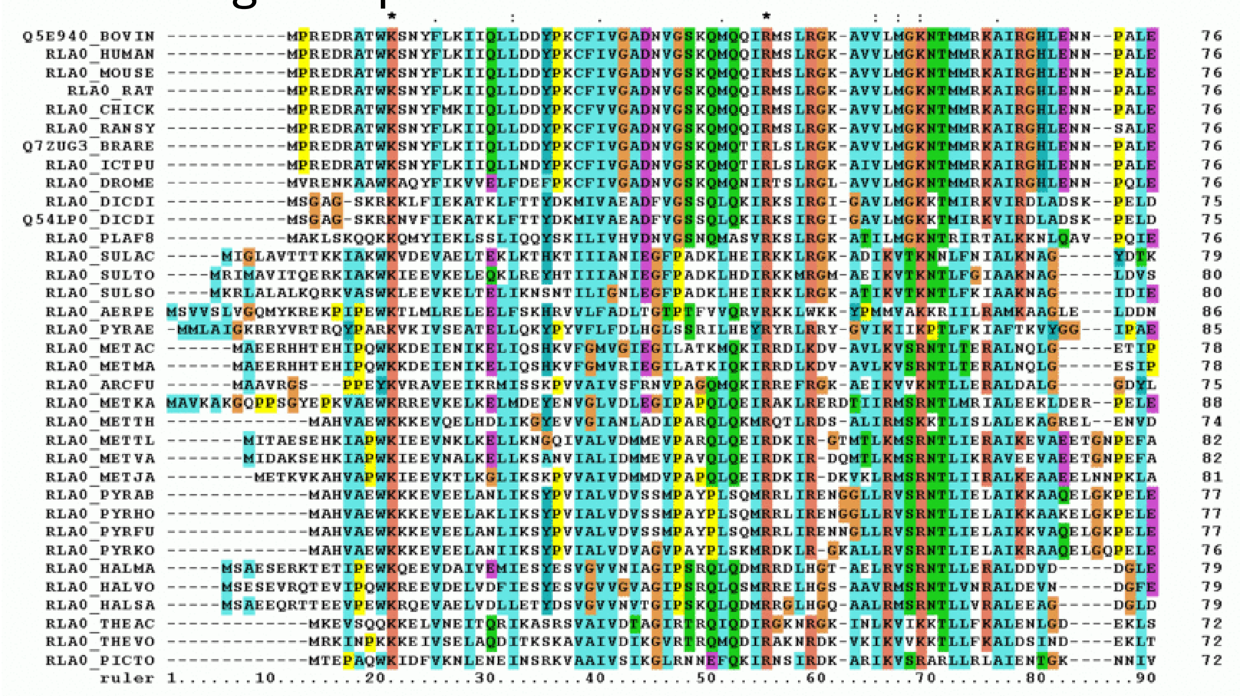
- PCs are linear combinations of data
- Outliers
- Missing data

PCA examples

See exercises 4-8

Hierarchical clustering groups data points in a nested fashion by similarity

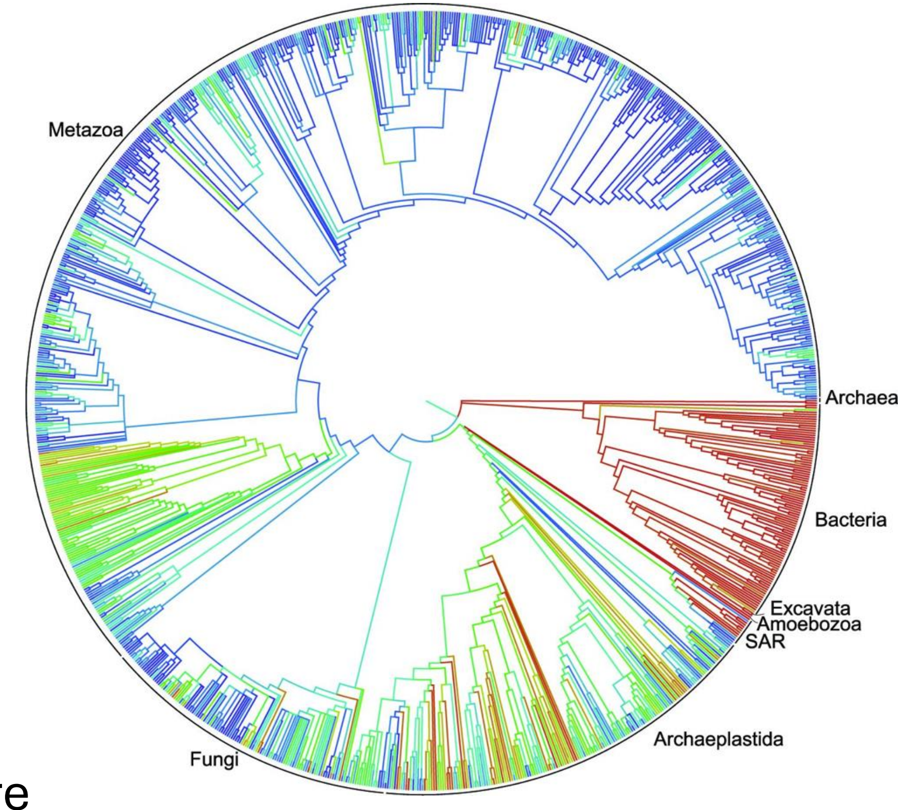
1. What data was used?
DNA sequences of biological species



https://en.wikipedia.org/wiki/Multiple_sequence_alignment

2. How was the distance estimated? A distance measure

Clustering of biological species



Hierarchical clustering groups data points in a nested fashion by similarity

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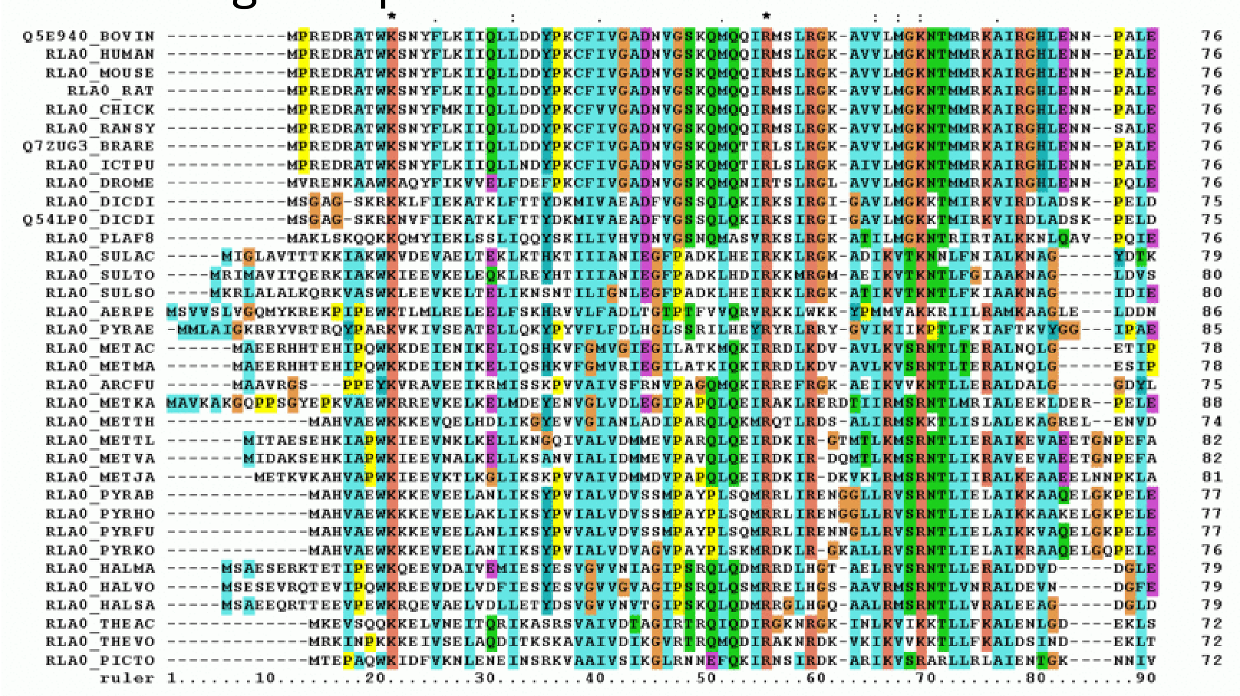
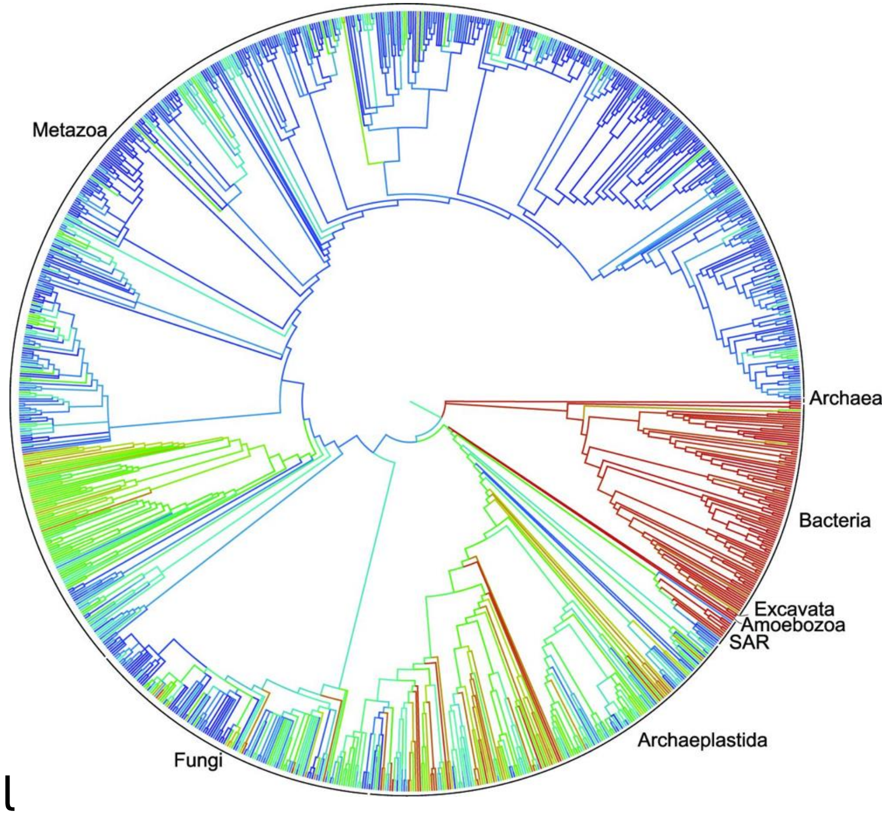
DNA sequences of biological species



https://en.wikipedia.org/wiki/Multiple_sequence_alignment

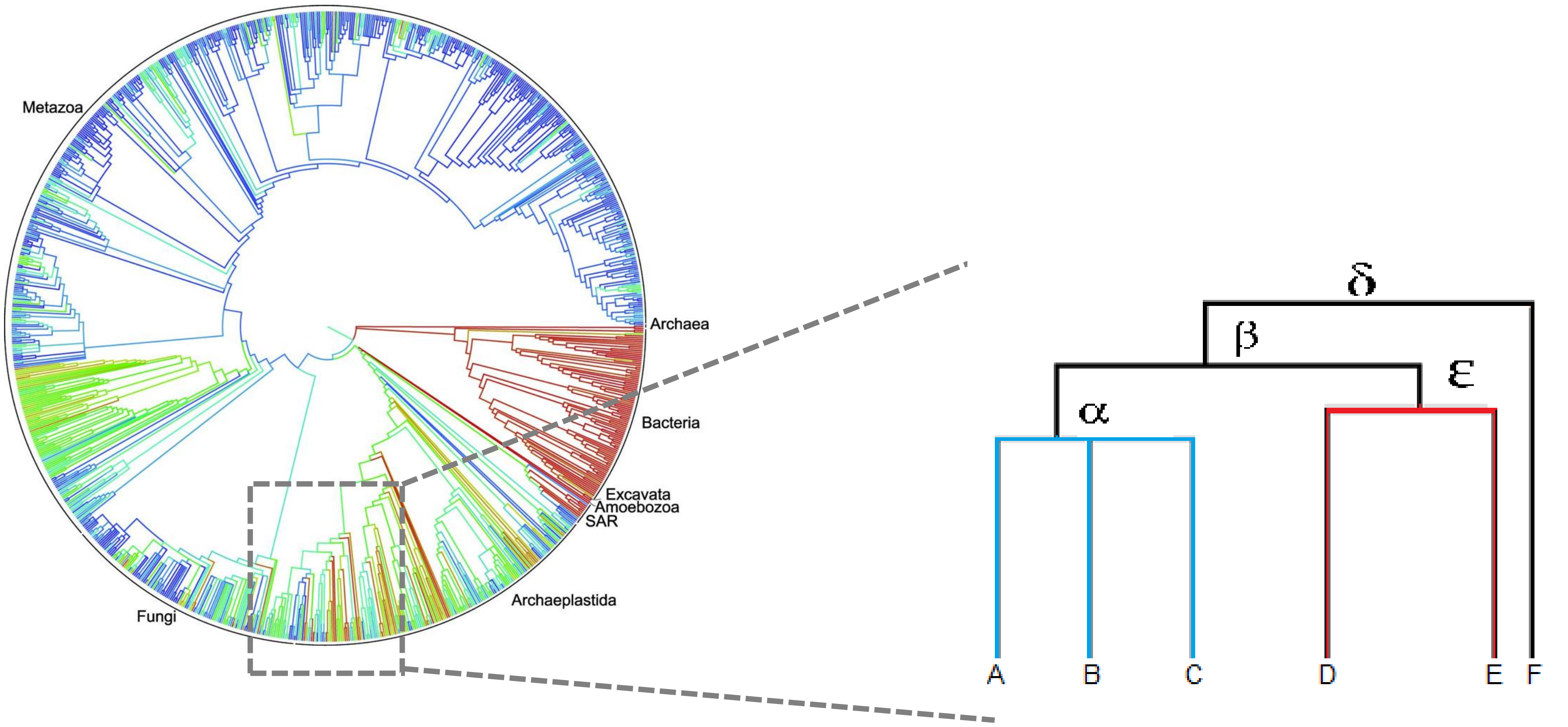
2. How was the distance estimated? An evolutional model

A phylogenetic tree



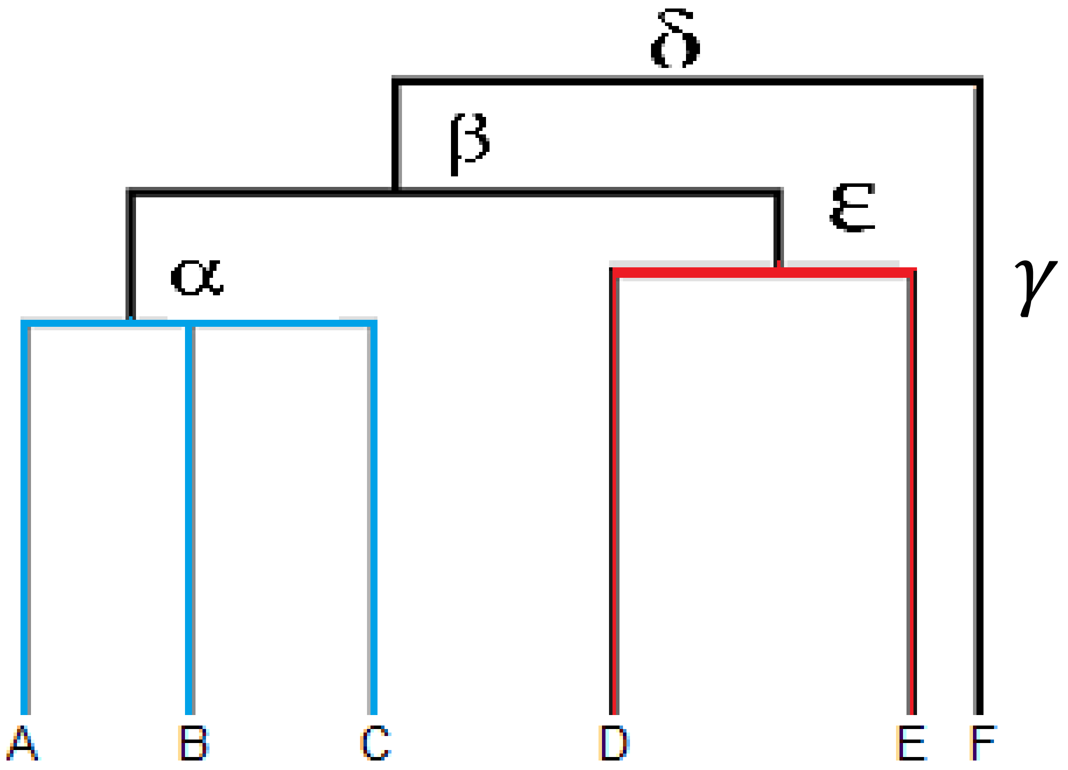
Hierarchical clustering makes a tree (dendrogram)

Clustering of biological species



Dendrogram visualizes hierarchical relationship of terminal nodes

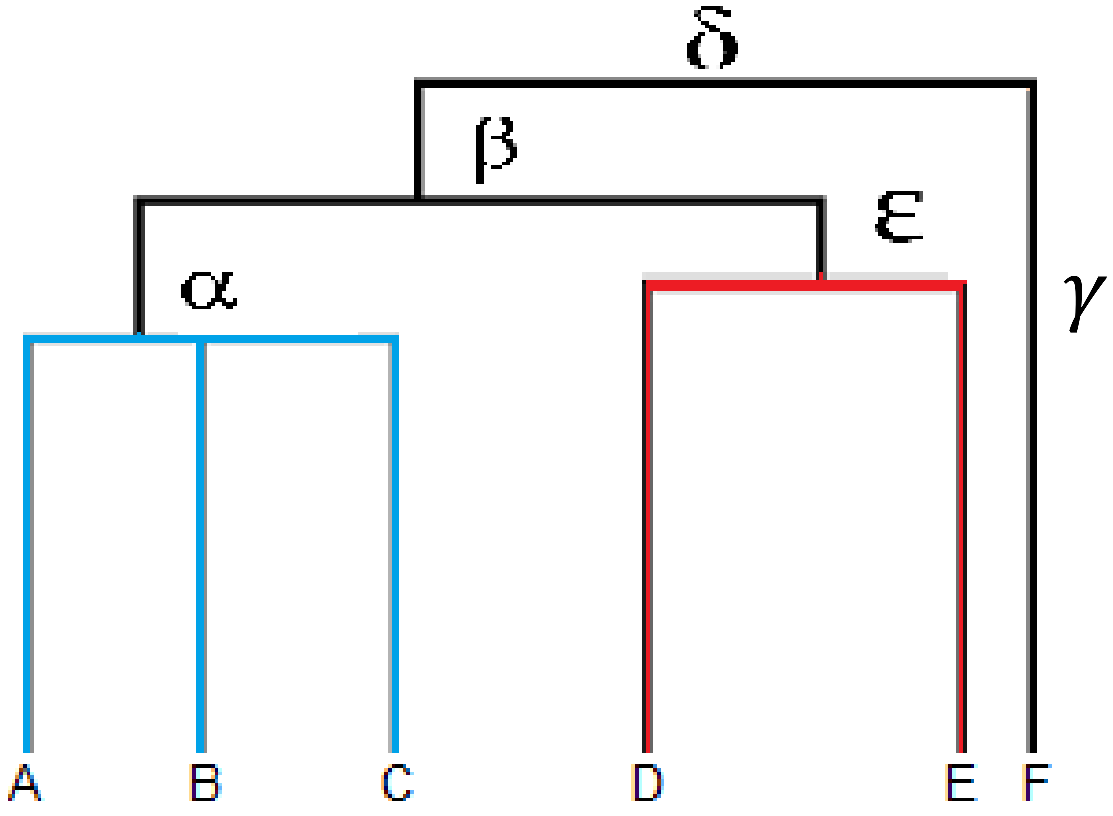
- Dendrogram = nodes + edges
- Nodes
  - root node: no parent allowed
  - intermediate nodes: multiple descendants
  - terminal nodes: data points
- Edges represent the relationship



https://www.statisticshowto.datasciencecentral.com/hierarchical-

Dendrogram visualize group memberships

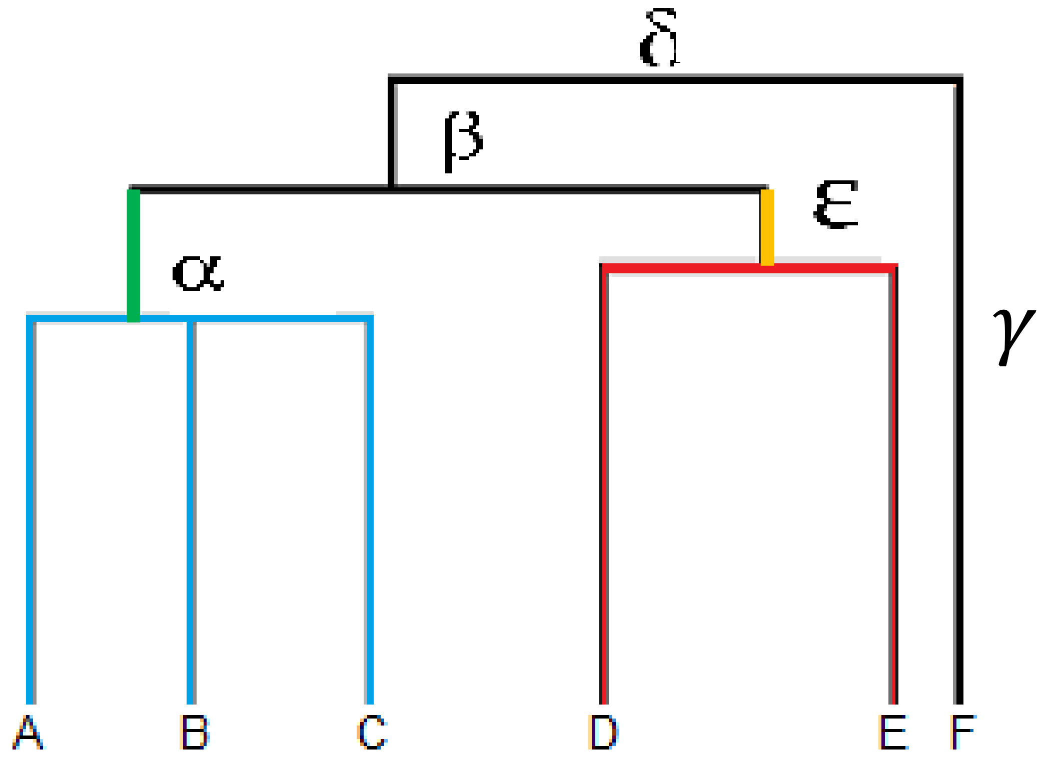
- Leaves A, B, and C are more similar to each other than to D, E, or F.
- Leaves D and E are more similar to each other than to A, B, C, or F.
- Leaf F is different from all of the other leaves.



https://www.statisticshowto.datasciencecentral.com/hierarchical-

Dendrogram visualizes distances among objects

- — indicates how A, B, C is different from each other
- — indicates how D, E is different from each other
- — (α) indicates how the clade of A, B, C is from the rest
- — (ϵ) indicates how the clade of D, E is different from the rest.
- — + — $(\alpha + \epsilon)$ shows how different the clade of A, B, C is from that of D, E

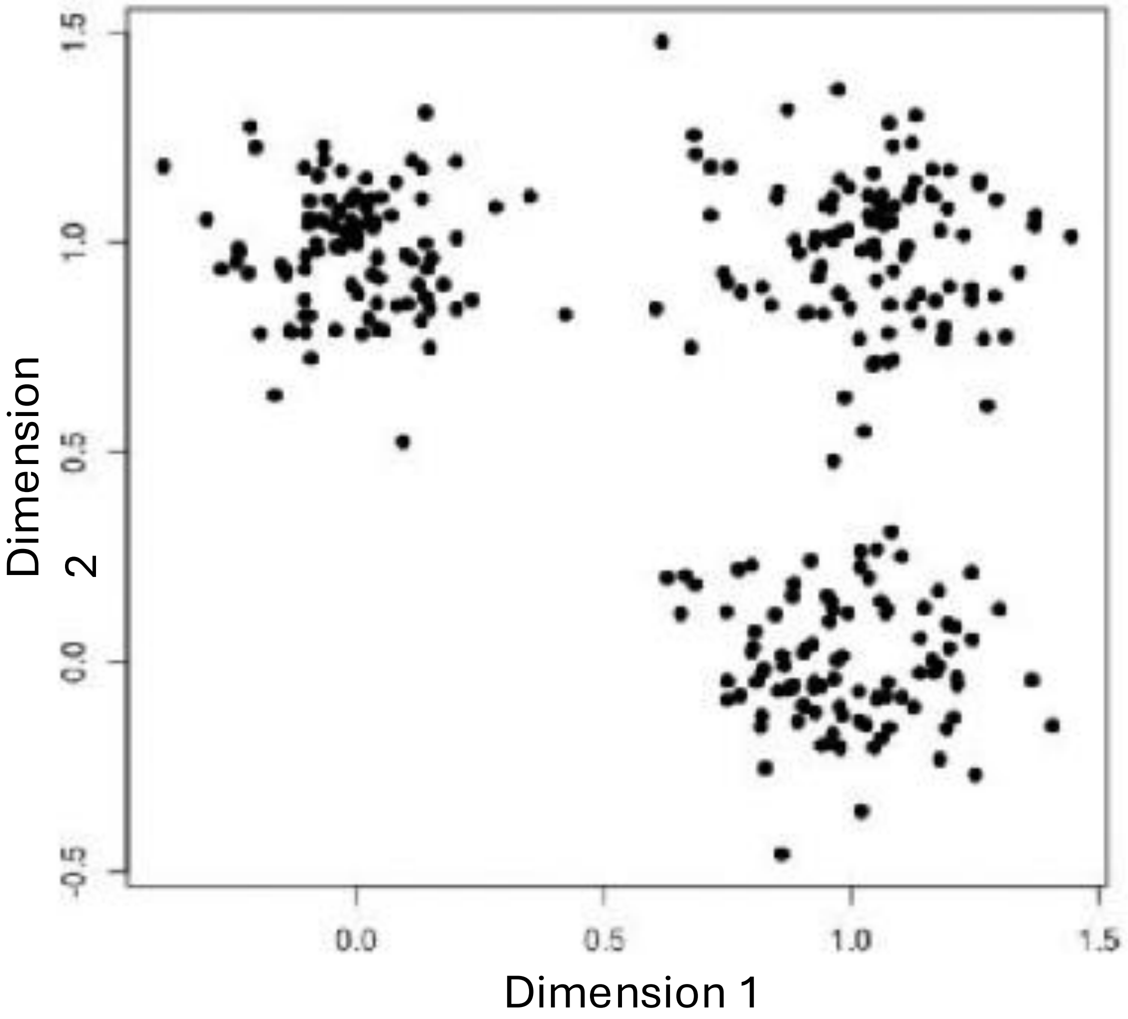


https://www.statisticshowto.datasciencecentral.com/hierarchical-

Clustering based on distance

Algorithm of K-means clustering on 2-D data

- Given data of **two** dimensions (features)

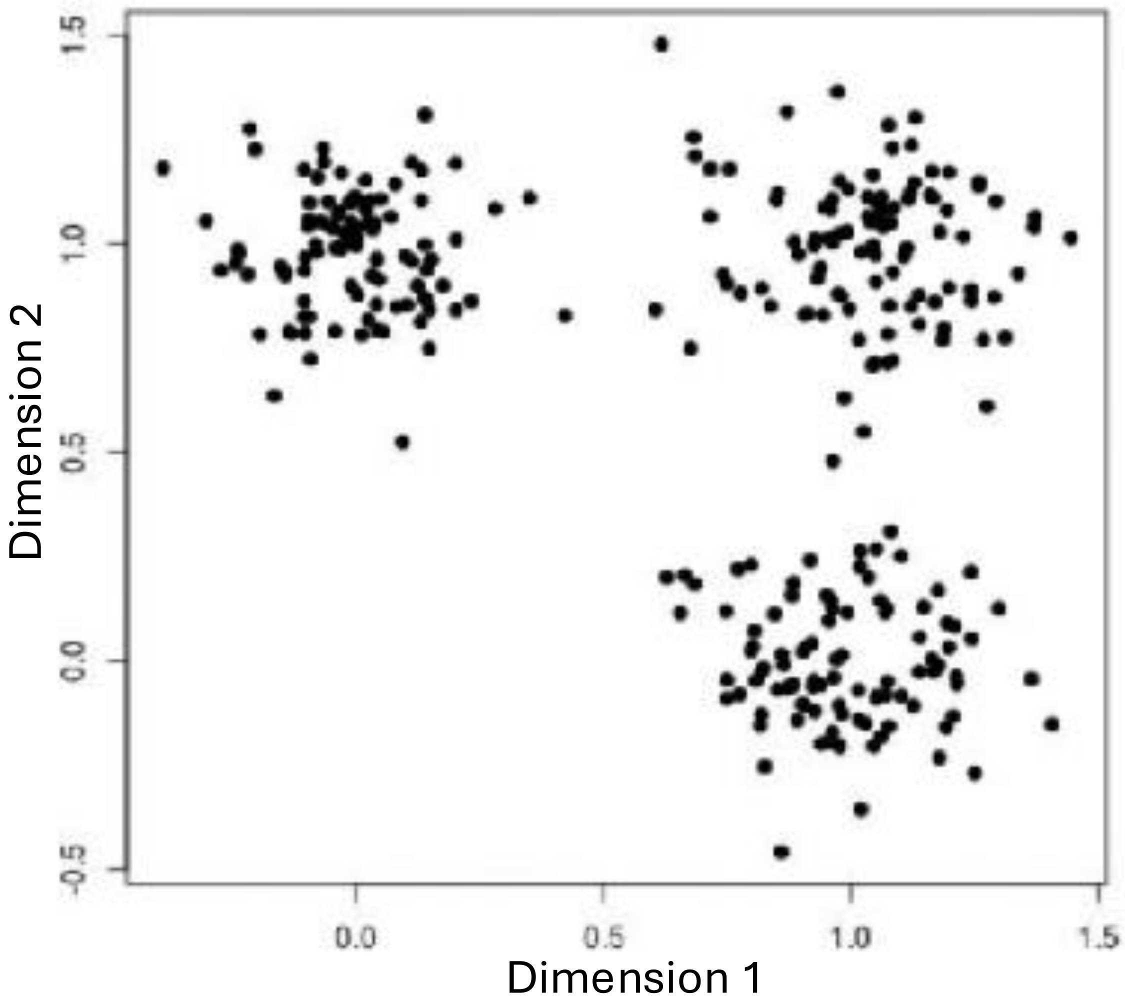


https://towardsdatascience.com/cluster-analysis-create-visualize-and-interpret-customer-segments-

Clustering based on distance

Algorithm of K-mean clustering on 2-D data

- Given data of **two** dimensions
- Assuming **three** clusters (K=3)

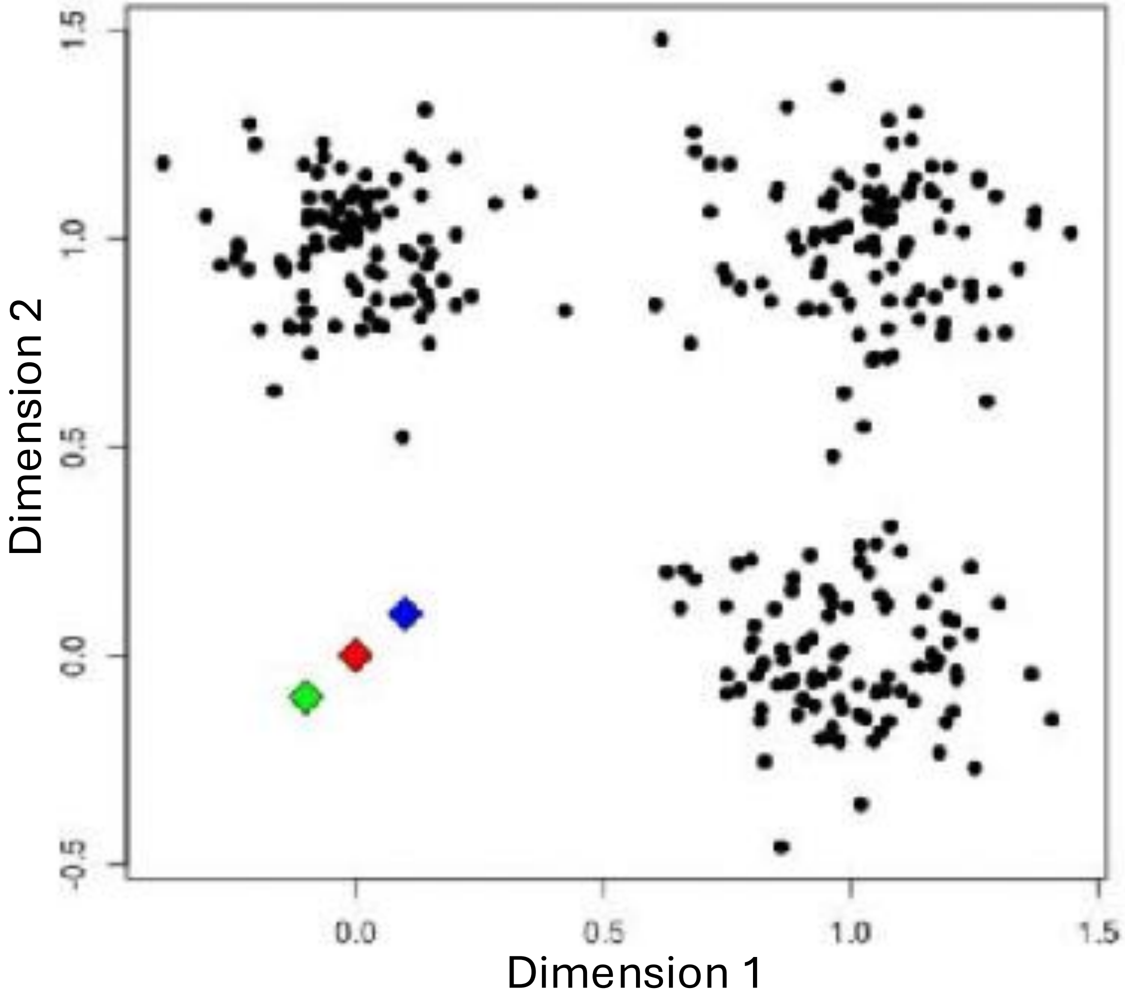


https://towardsdatascience.com/cluster-analysis-create-visualize-and-interpret-customer-segments-

Clustering based on distance

Algorithm of K-mean clustering on 2-D data

- Given data of **two** dimensions
- Assuming **three** clusters (K=3)
- **Randomly** choose the center point (centroid) for the clusters

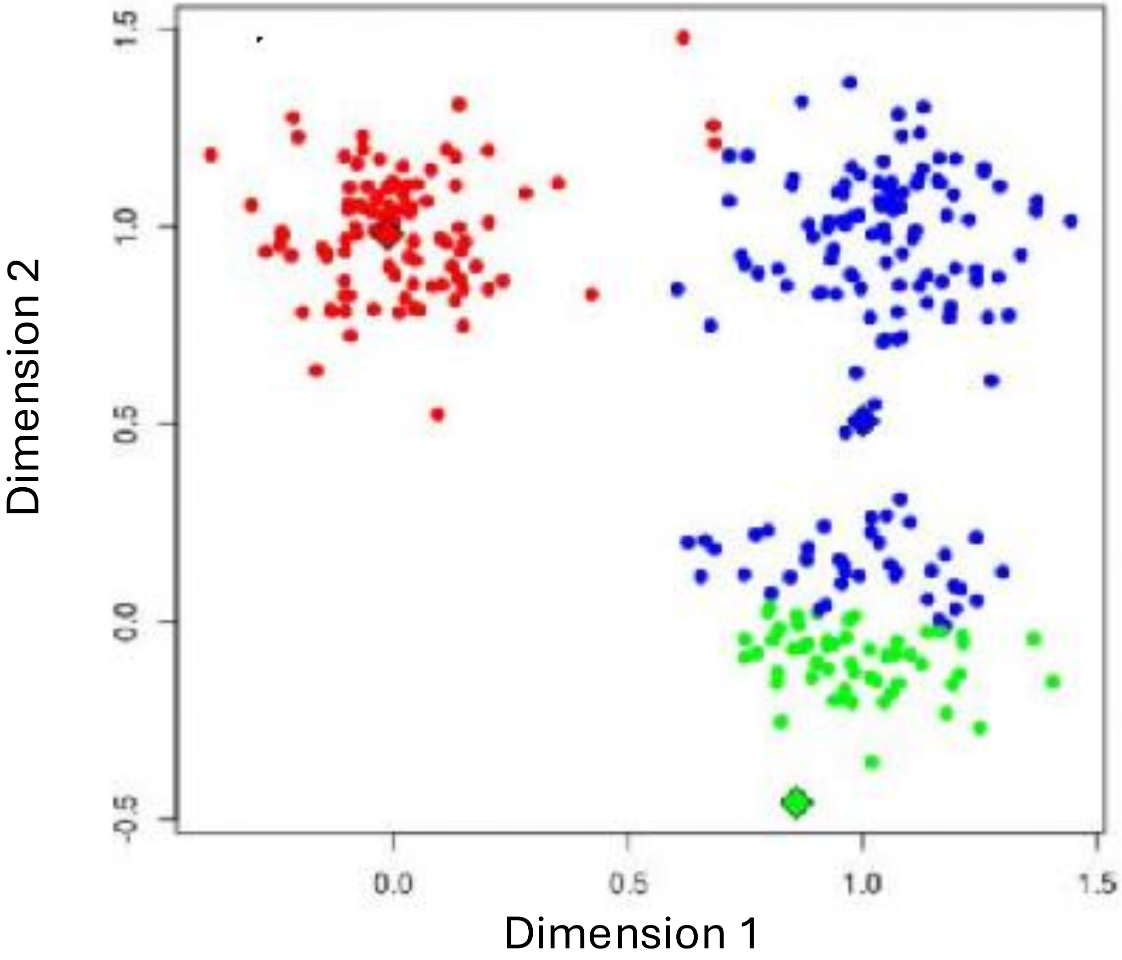


https://towardsdatascience.com/cluster-analysis-create-visualize-and-interpret-customer-segments-

Clustering based on distance

Algorithm of K-mean clustering on 2-D data

- Given data of **two** dimensions
- Assuming **three** clusters (K=3)
- Update cluster assignments based on the **distance**

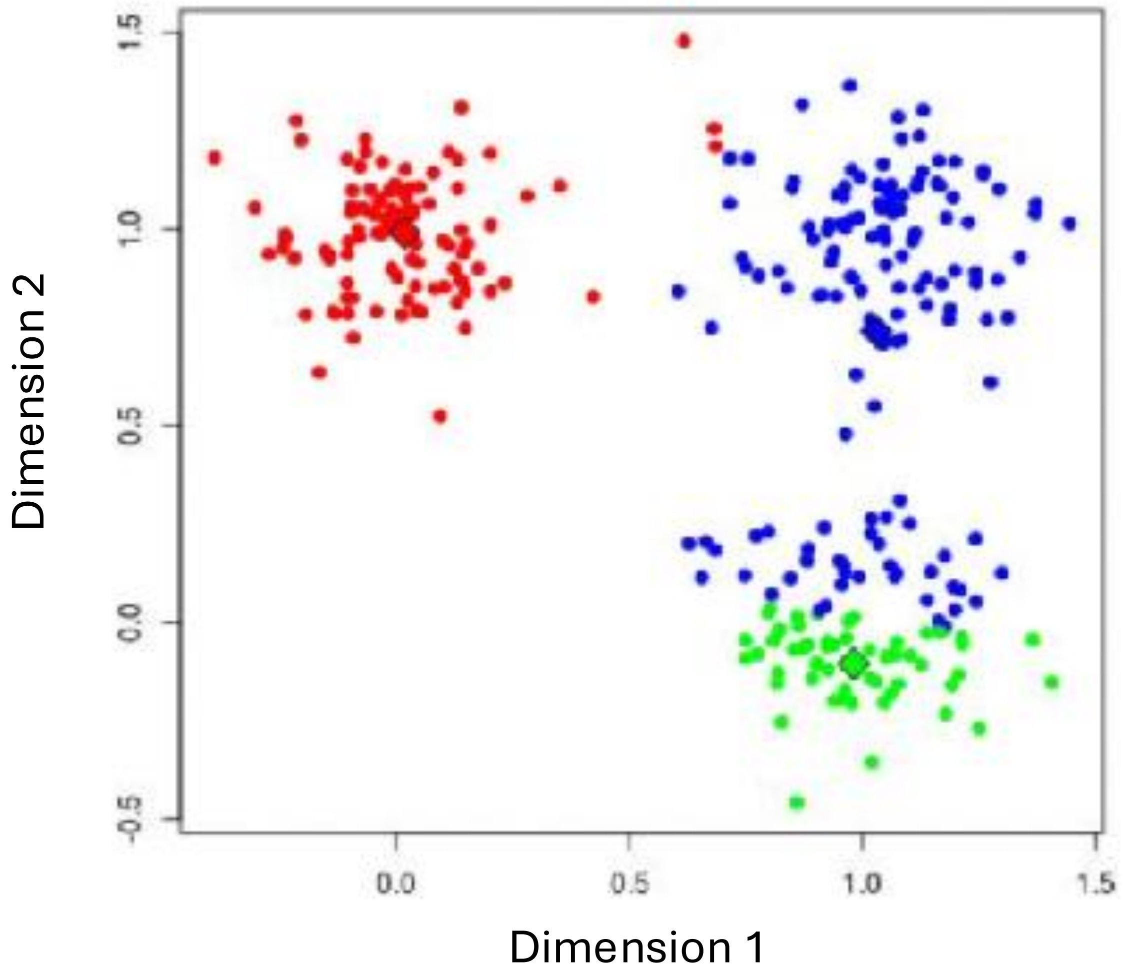


https://towardsdatascience.com/cluster-analysis-create-visualize-and-interpret-customer-segments-

Clustering based on distance

Algorithm of K-mean clustering on 2-D data

- Given data of **two** dimensions
- Assuming **three** clusters (K=3)
- Update the cluster centers



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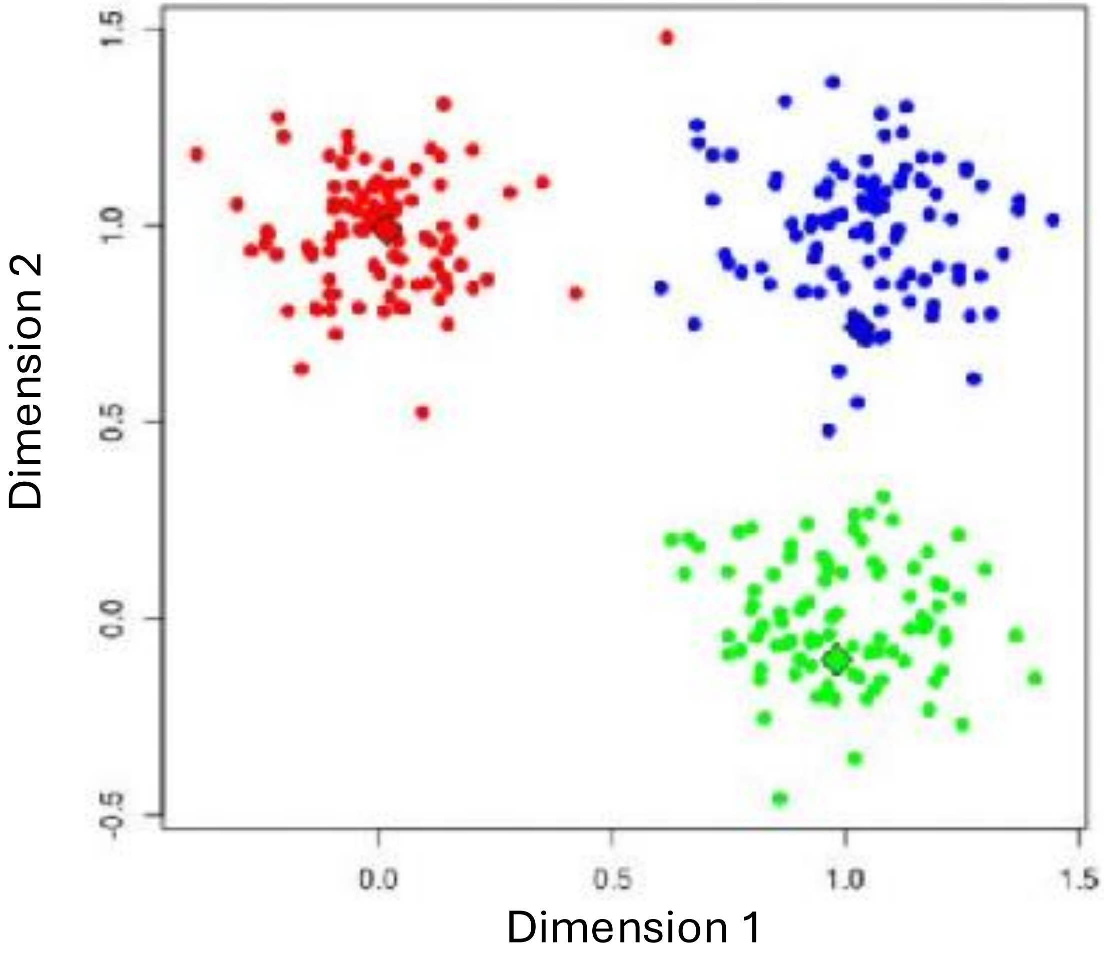
Clustering based on distance

Algorithm of K-means clustering on 2-D data

- Given data of **two** dimensions
- Assuming **three** clusters (K=3)
- Update cluster assignments based on the **distance**

Things to control

- How to determine K
- Location of the initial points
- When to stop
- **How to calculate distance**



https://towardsdatascience.com/cluster-analysis-create-visualize-and-interpret-customer-segments-

Dimension reduction and clustering example

See exercises 9-10

How “unsupervised”?

The models don’t decide everything for you!

- Pre-process the data?
- Feature selection?
- Linear or non-linear?
- What distance metric? (What weights?)
- How many clusters?
- Etc.